HOMEWORK 2.1 · UNIT 2 · NC.8.EE.1

Introduction to Exponents

Targets: EE.1-1, EE.1-7, EE.1-5, EE.1-8, EE.1-9

Name _____ Date _____ Block _____

Remember: A **power** has a **base** and an **exponent**. The exponent tells how many times the base is used as a **factor**. Example: $5^3 = 5 \cdot 5 \cdot 5 = 125$ (not 5×3). Parentheses matter: $(-3)^2 = 9$, but $-3^2 = -9$.

SCORE 3.0 Approaching · the basics

Targets: EE.1-1

1 Name the base and the exponent. 5^8 EE.1-1

base, exponent: _____

2 Write 3^5 in expanded form. Then find the value. EE.1-1

_____

_____

3 Write with exponents. $a \cdot a \cdot a \cdot a \cdot b \cdot b$ EE.1-1

_____

4 What is the value of 3^4? EE.1-1

(A) 12 (B) 7 (C) 81 (D) 64

5 4^3 means 4×3. EE.1-1

True False

SCORE 4.0 Proficient · on target

Targets: EE.1-7, EE.1-1, EE.1-5

6 **Priya** made a mistake. Circle it. Explain it. Fix it. EE.1-7

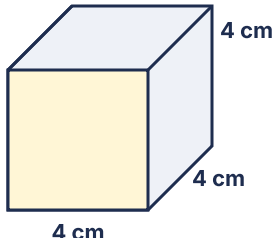
PRIYA'S WORK
$(-5)^2 = -25$

7 Which are equal to **81**? EE.1-1,EE.1-5

☐ 3^4 ☐ 9^2 ☐ 8^1 ☐ 4^3
☐ $(-9)^2$

Select all that apply.

8 The cube has edges 4 cm long. Which shows its volume? EE.1-1



Ⓐ $4^3 = 64$ cm^3
Ⓑ $4 \times 3 = 12$ cm^3
Ⓒ $4^2 = 16$ cm^3
Ⓓ $3^4 = 81$ cm^3

SCORE 5.0 Mastery · go deeper

Targets: EE.1-8, EE.1-9

9 Is -3^2 the same as $(-3)^2$? Explain. EE.1-8

No, because -3^2 = _____ and $(-3)^2$ = _____ .

10 Jay says 10^6 is a 1 followed by 6 zeros. Is he right? Explain. EE.1-9

Jay is _____ because _____ .